Causal Inference Basics

(Causal Identification with Directed Acyclic Graphs)

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11:10am

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Why DAGs are worth knowing

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DAGs formalize when (1) and not (2).

Learning goals for today

- fork structures
- collider structures
- causal reasoning and statistical independence

A hypothetical experiment in two population subgroups

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People who like exercise

People who don't like exercise

A hypothetical experiment in two population subgroups

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75% assigned an exercise coach for 1 month

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Outcome: How many pull-ups can they do?

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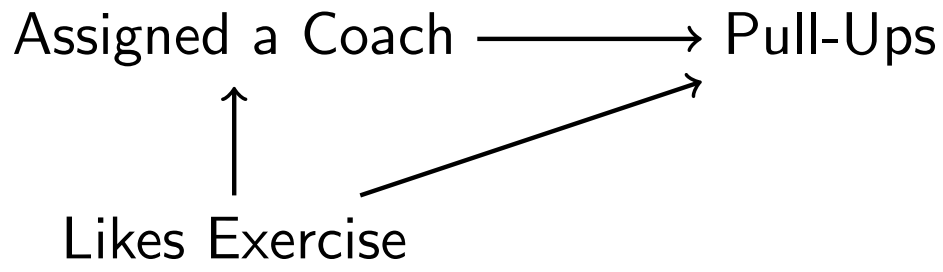
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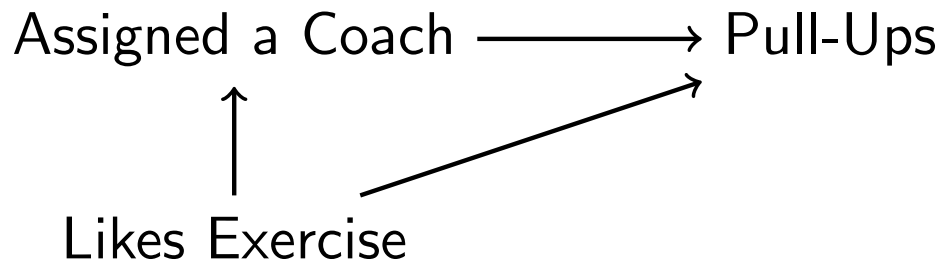
Outcome: How many pull-ups can they do?

Question for you:
Give 2 reasons why those assigned a coach can do more pull-ups

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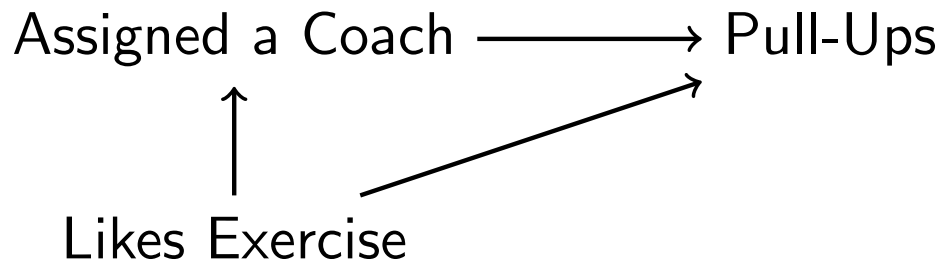


A hypothetical experiment in two population subgroups



Nodes are random variables. **Edges** ($\rightarrow$) are causal relations

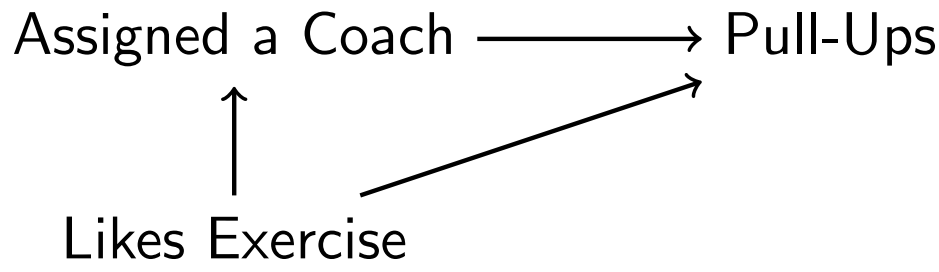
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The graph links causal assumptions to statistical dependence

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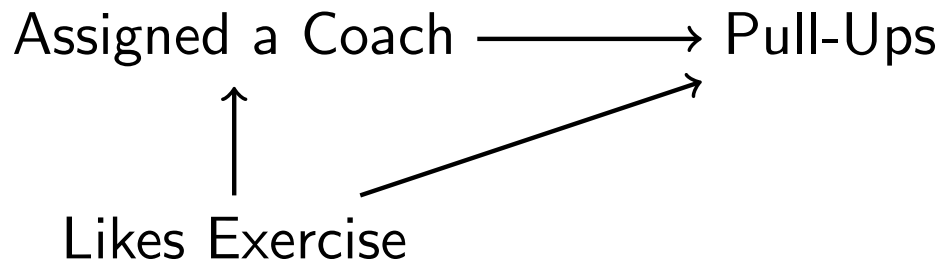


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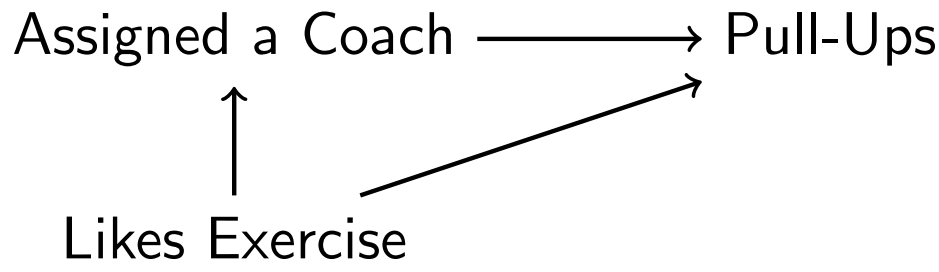
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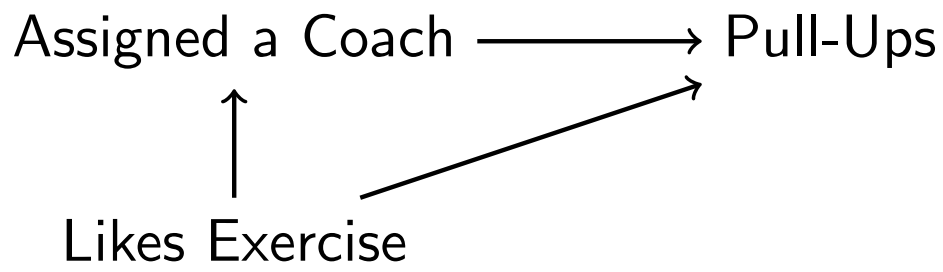
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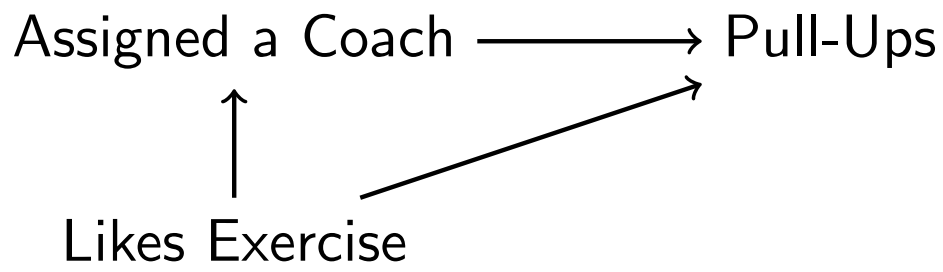
- (Assigned a Coach) $\rightarrow$ (Pull-Ups)
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- (Assigned a Coach) $\leftarrow$ (Likes Exercise) $\rightarrow$ (Pull-Ups)
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How to study the causal effect (Assigned a Coach) $\rightarrow$ (Pull-Ups)?

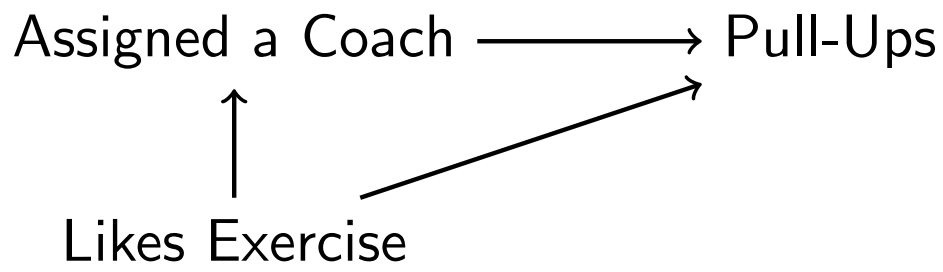
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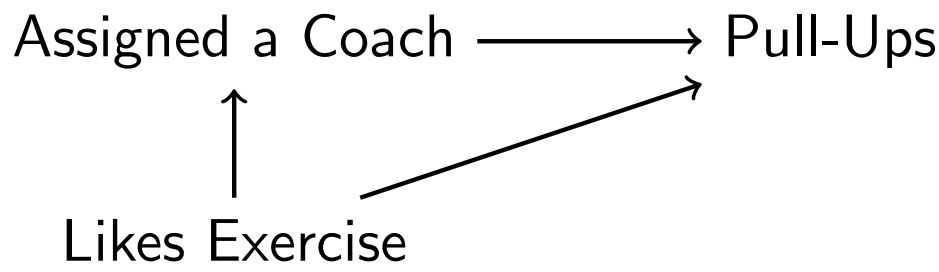
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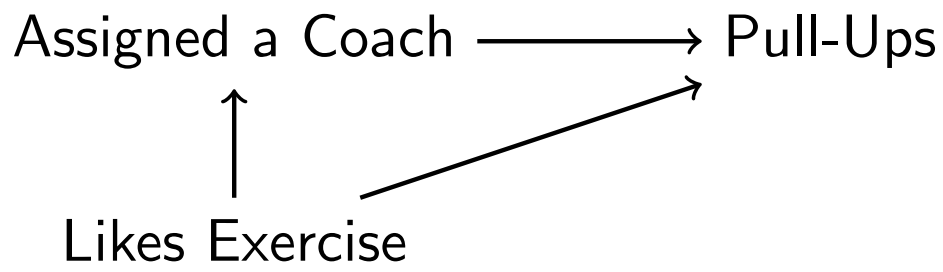
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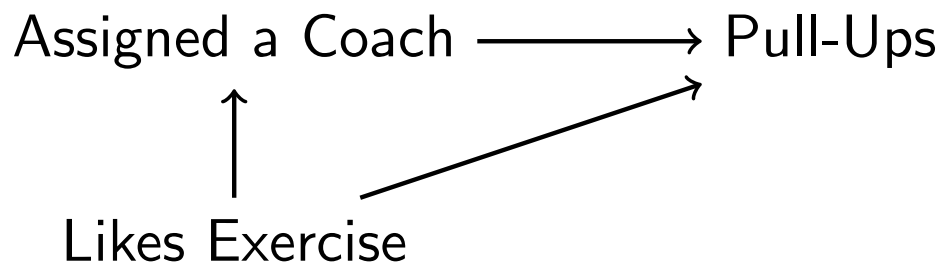
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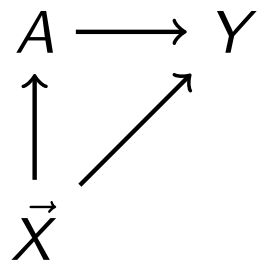
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Terminology: We condition on [Likes Exercise]

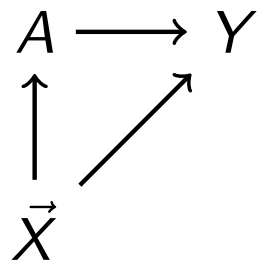
Recap: Why DAGs are worth knowing

1. DAGs tell us when conditioning on $\vec{X}$ identifies a causal effect



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DAGs also tell us when conditioning can create problems.

Colliders: The sprinkler example

Example from Pearl, J. (1988). Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference.

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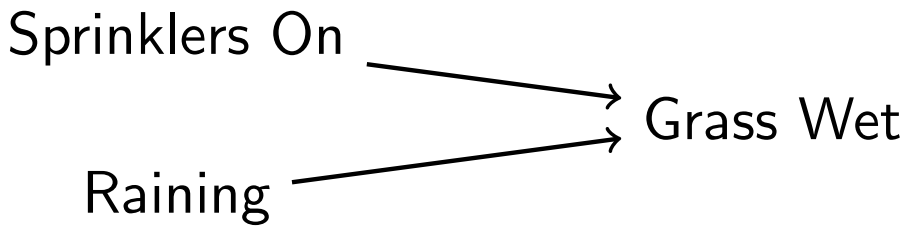
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- I set my sprinklers to turn on at random times
- It rains at random times
- (Sprinklers) or (Rain) can make the grass wet

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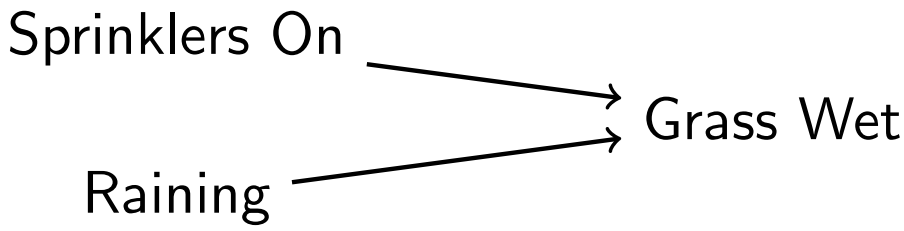
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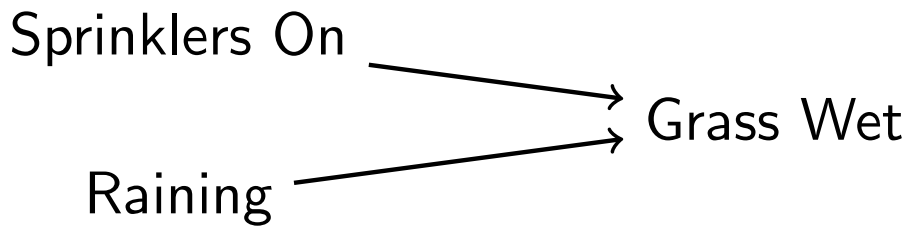


Questions:

- If (Sprinklers On = FALSE), does that help me predict (Raining)?
- If (Sprinklers On = FALSE) and (Grass Wet = TRUE), does that help me predict (Raining)?

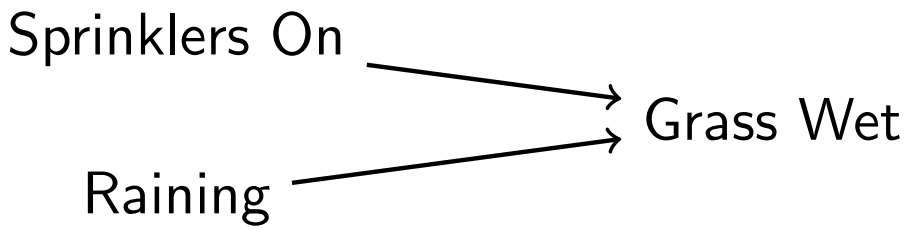
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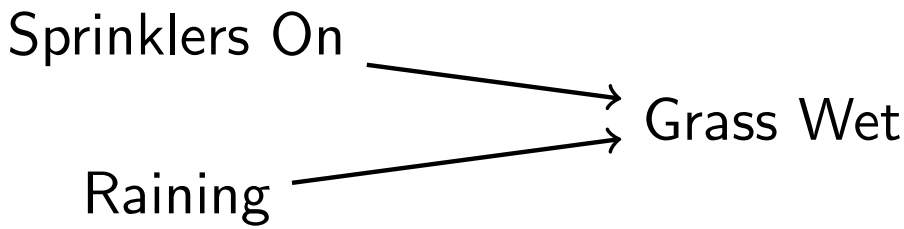
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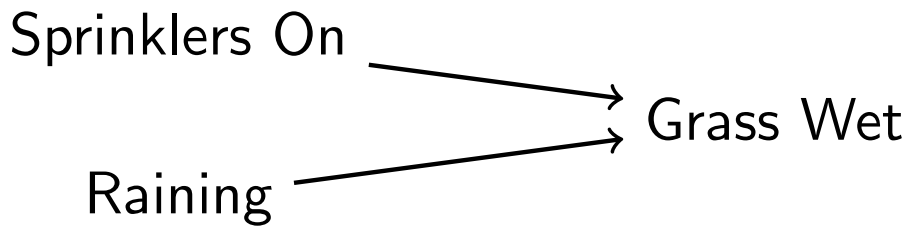
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- A collider blocks a path
  - marginal independence of (Sprinklers On) and (Raining)

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  - marginal independence of (Sprinklers On) and (Raining)
- Conditioning on a collider opens the path
  - conditional dependence of (Sprinklers On) and (Raining) when restricting to times when (Grass Wet = `True`)

Using DAGs to identify causal effects: Game plan

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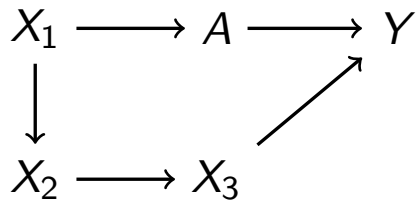
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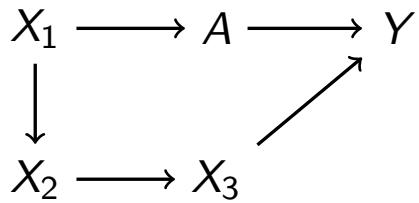
Exercise 1

Find adjustment sets that identify the effect of A on Y



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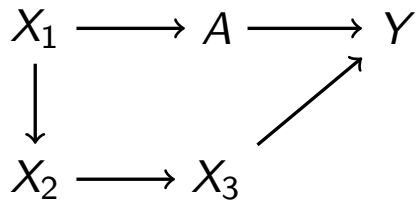
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We can block the backdoor path in several ways:

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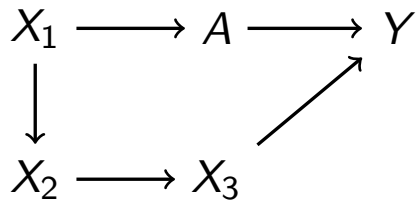


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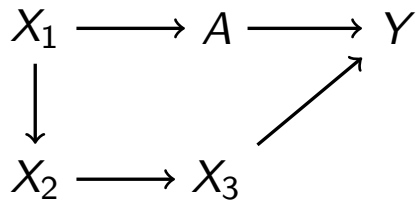


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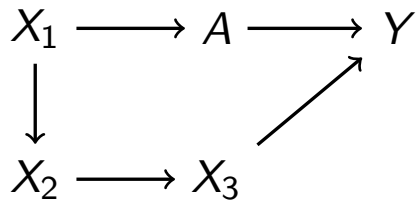


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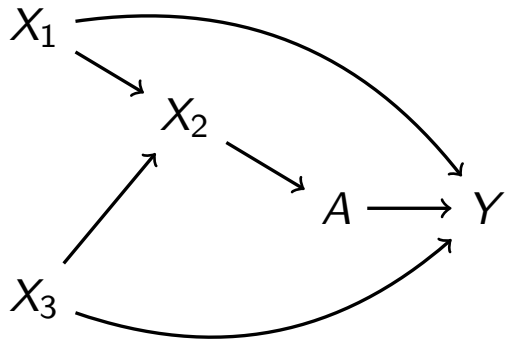


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- Any combination of the above

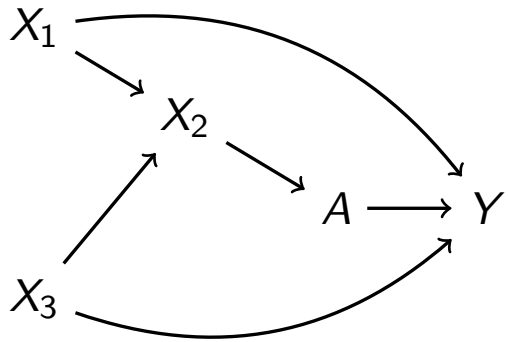
Exercise 2

Find 3 sufficient adjustment sets to identify $A \rightarrow Y$



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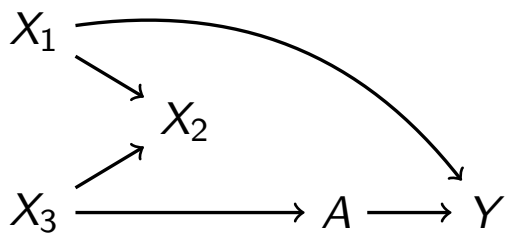
Find 3 sufficient adjustment sets to identify $A \to Y$



Answer: $\{X_2\}, \{X_1, X_3\}, \{X_1, X_2, X_3\}$

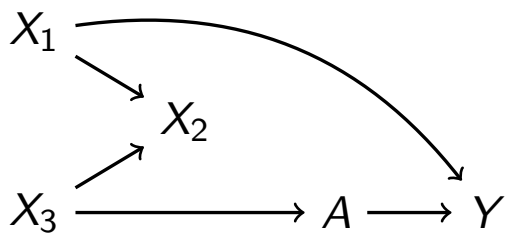
Exercise 3

What is the smallest adjustment set that identifies $A \to Y$?



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What is the smallest adjustment set that identifies $A \to Y$?



Answer: The empty set! Don't condition on anything.
The collider X_2 already blocks the path.

DAG in a realistic setting

To what extent does completing a 4-year college degree affect a person's future earnings?

Effect of a 4-year degree on future earnings

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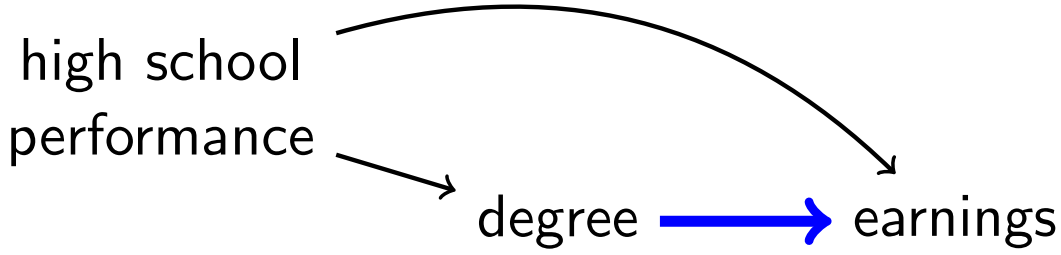
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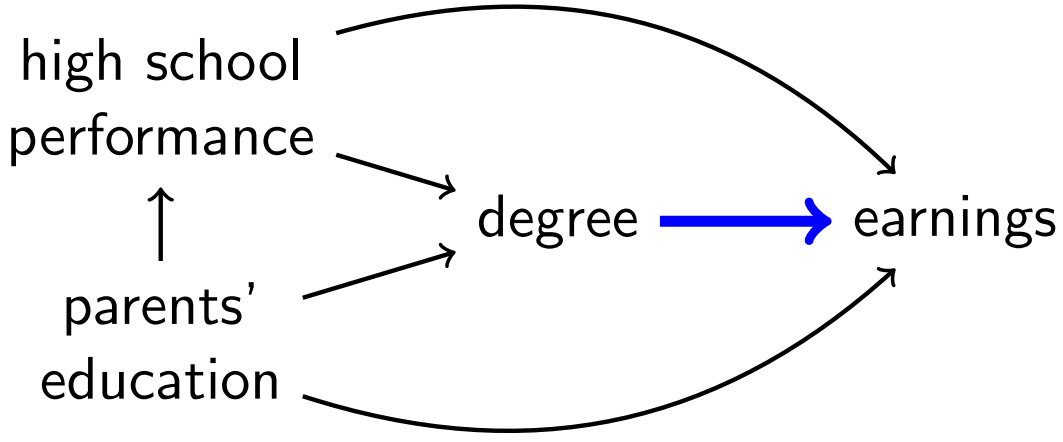
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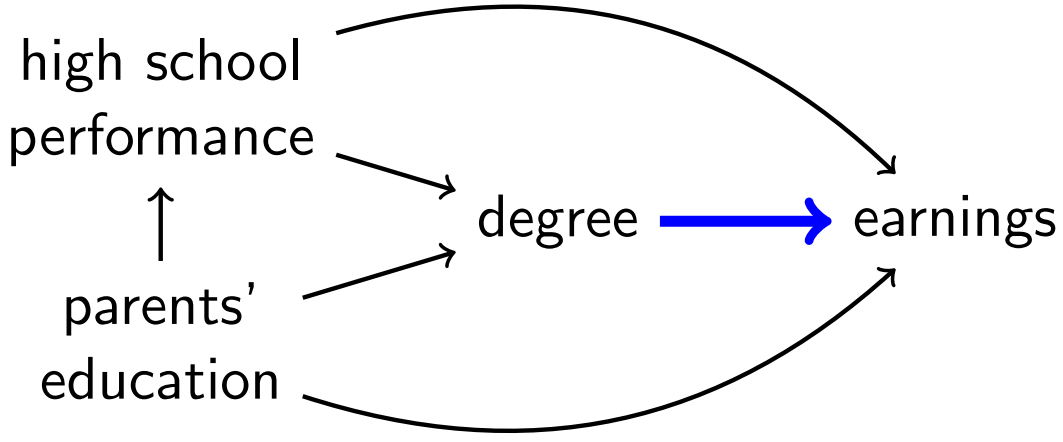
Effect of a 4-year degree on future earnings

1) Draw a DAG



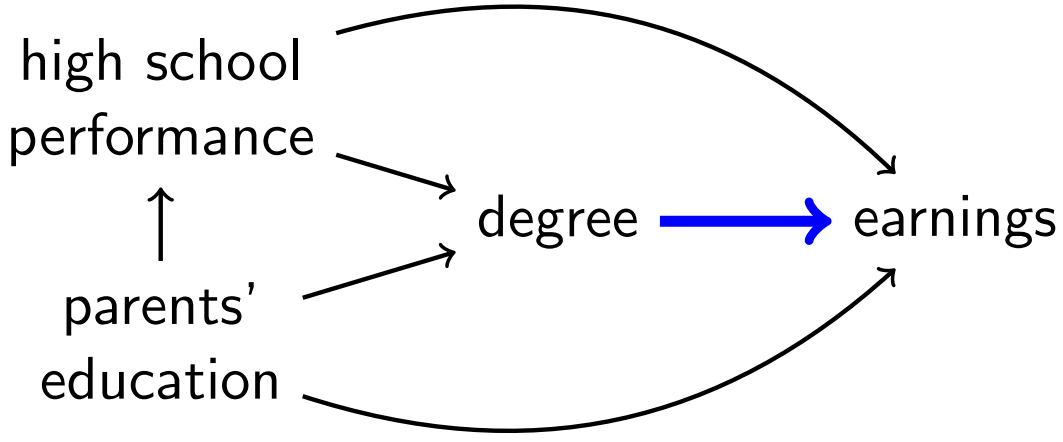
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2) List all paths between the treatment and outcome



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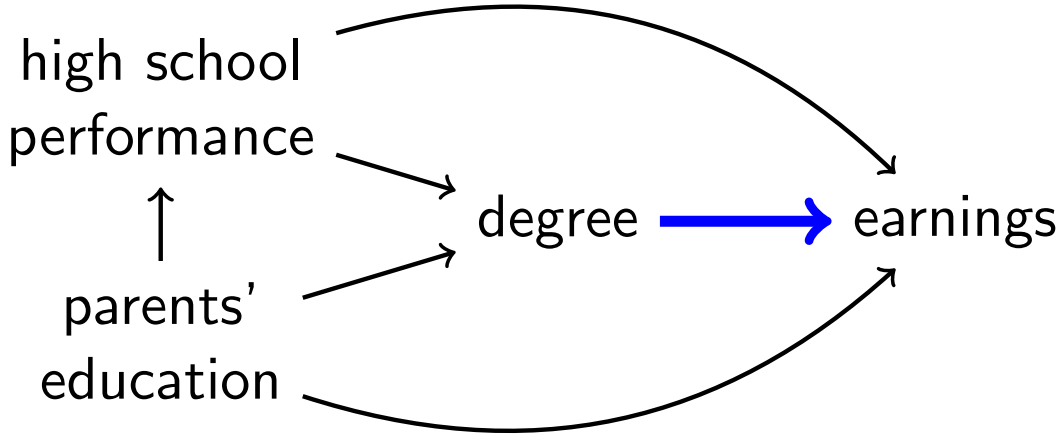


Causal paths

(degree) $\rightarrow$ (earnings)

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Causal paths

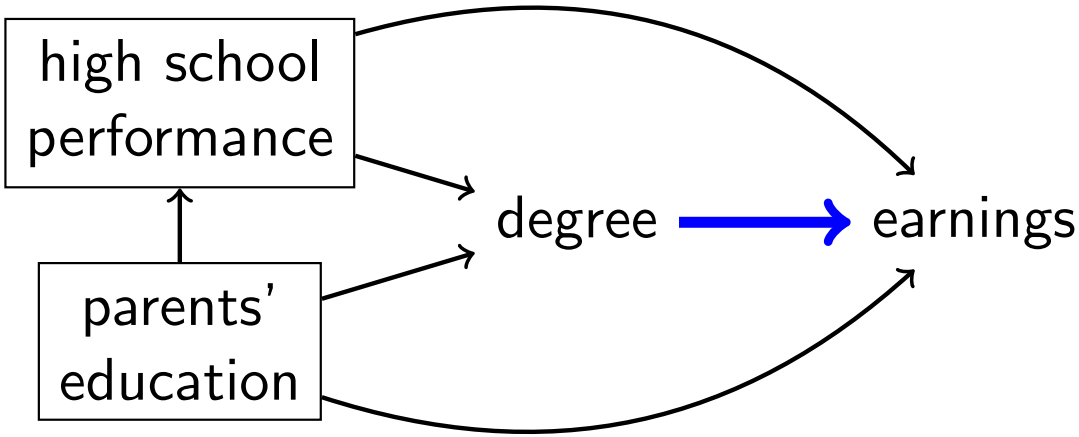
(degree) $\rightarrow$ (earnings)

Backdoor paths

(degree) $\leftarrow$ (high school performance) $\rightarrow$ (earnings)
(degree) $\leftarrow$ (parents' education) $\rightarrow$ (earnings)
(degree) $\leftarrow$ (high school performance) $\leftarrow$ (parents' education) $\rightarrow$ (earnings)

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3) Choose a sufficient adjustment set
{high school performance, parents' education}



Causal paths

(degree) → (earnings)

Backdoor paths

(degree) ← high school performance → (earnings)

(degree) ← parents' education → (earnings)

(degree) ← high school performance ← parents' education → (earnings)

DAGs: A promising path

- DAGs connect causal theories to statistical dependence
- Statistical dependence arises through causal paths
- Paths may contain two key structures
  - forks: $A \leftarrow B \rightarrow C$
    (A and C dependent if B unadjusted)
  - colliders: $A \rightarrow B \leftarrow C$
    (A and C dependent if B adjusted)
- Causal identification goal:
  choose a sufficient adjustment set so only the causal path of interest remains open
- Experimental analog:
  Among units who are identical on the sufficient adjustment set, we have a simple randomized experiment

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  If this is the DAG,
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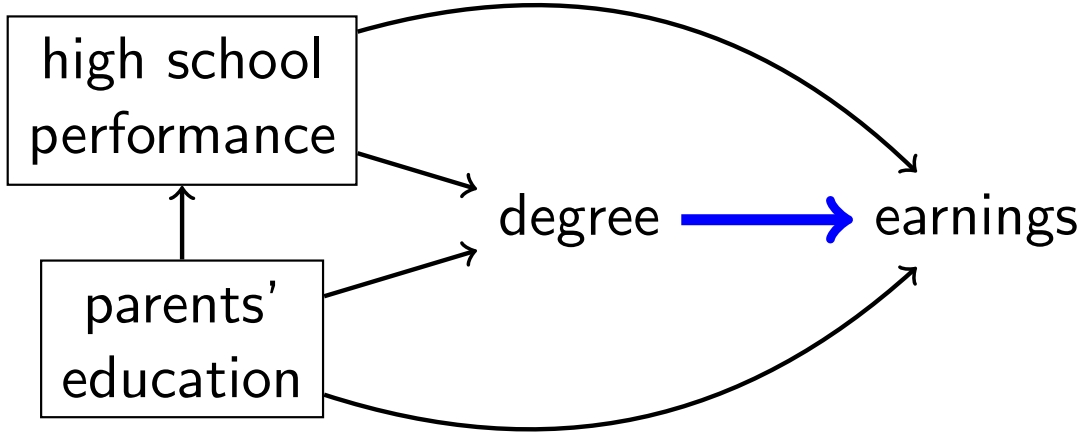
It is important to reason about when the DAG may not hold

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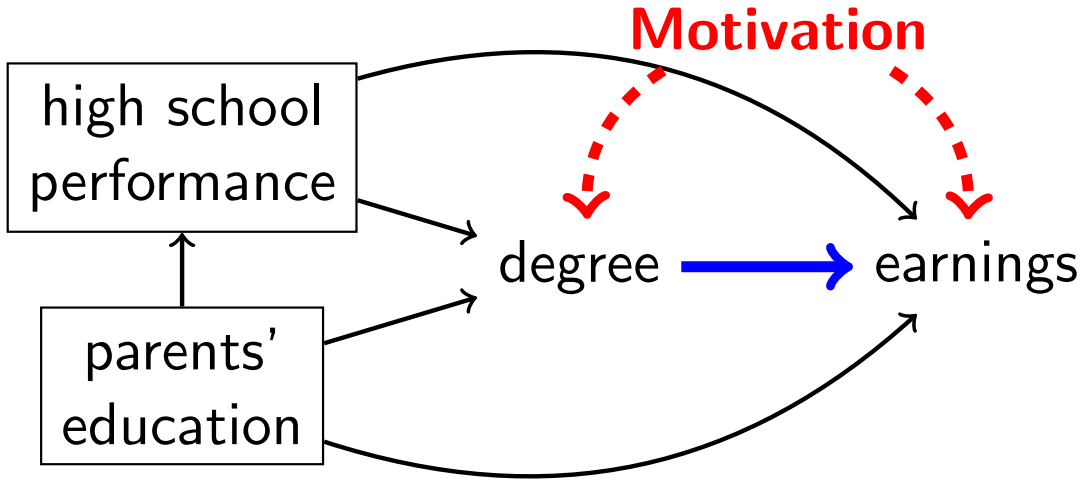


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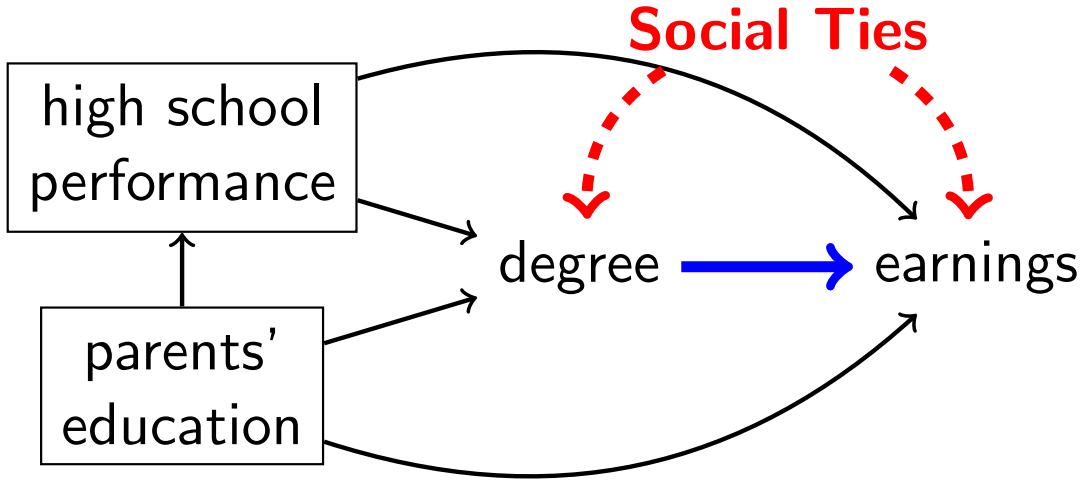


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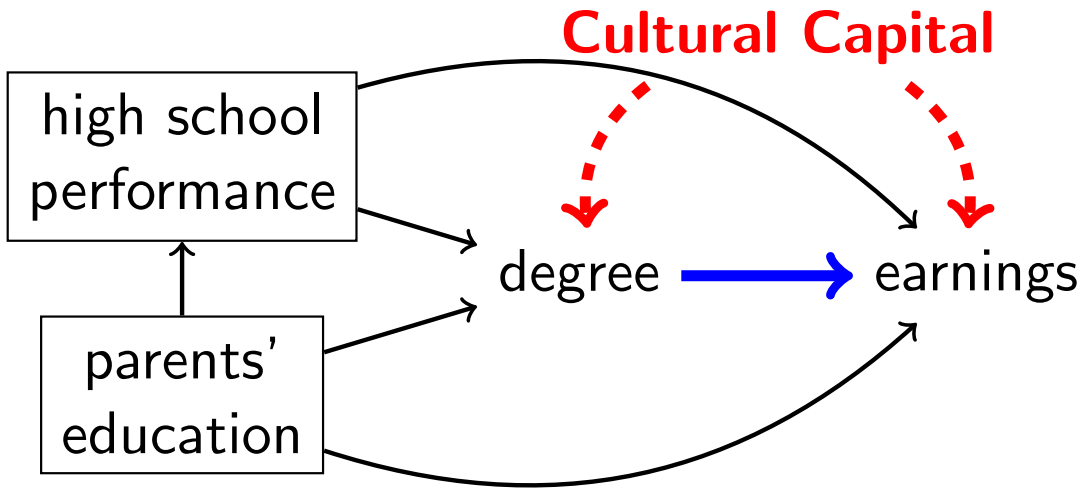


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Resources to learn more

- Hernán, M.A., & J.M. Robins. 2020.
  Causal Inference: What If?
  Boca Raton: Chapman & Hall / CRC.
- Pearl, J., & Mackenzie, D. (2018).
  The Book of Why: The New Science of Cause and Effect.
  Basic Books.
- Pearl, J., Glymour, M., & Jewell, N. P. (2016).
  Causal Inference in Statistics: A Primer.
  John Wiley & Sons.
- Pearl, J. (2000).
  Causality.
  Cambridge University Press.